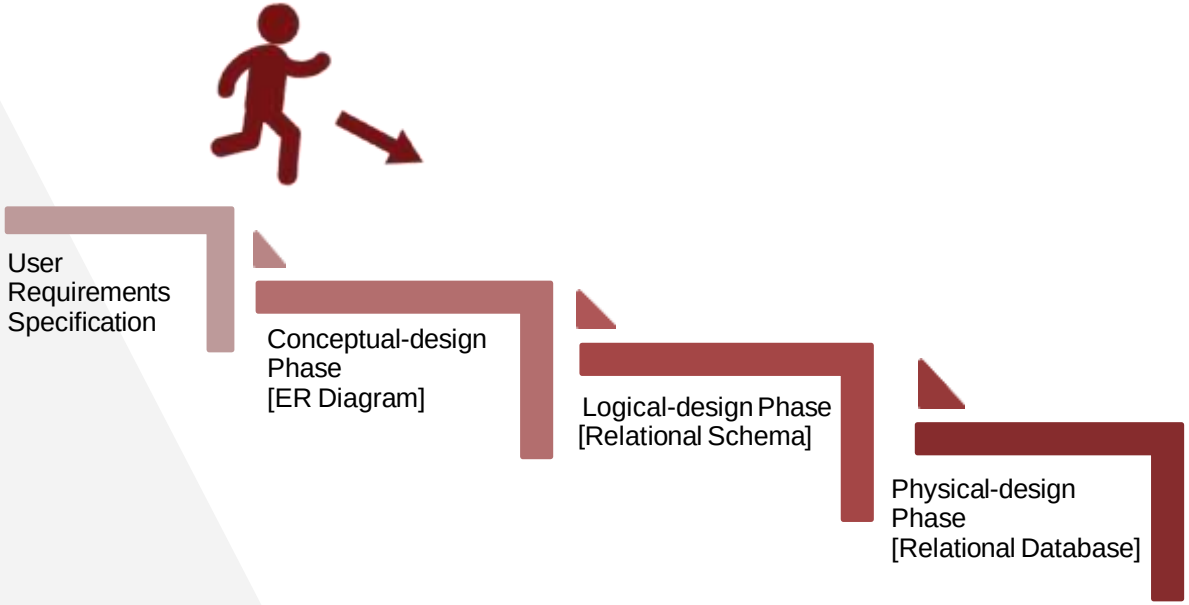


Database Systems

E-R Data Model

Design Phases



Conceptual Design ≫E-R Data Model

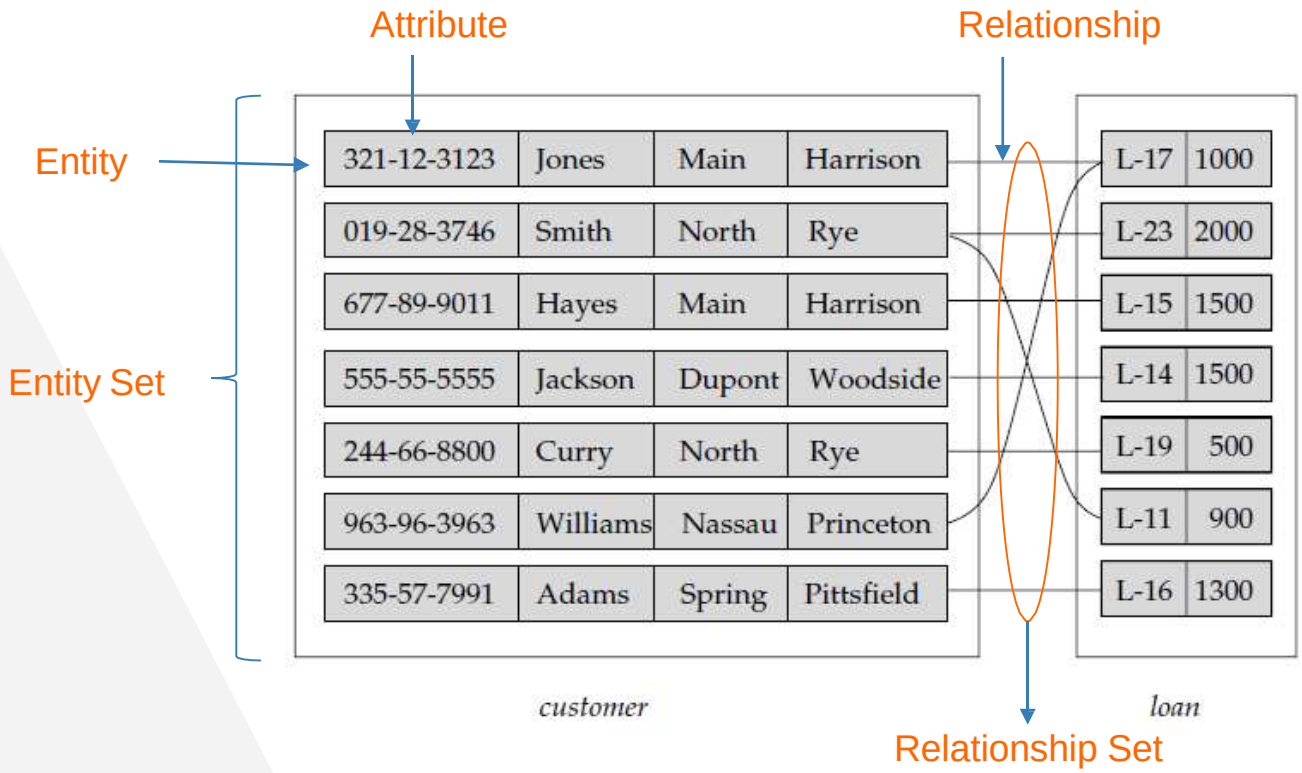
- Entity-relationship(E-R) data model is used to represent the conceptual design. It maps the meanings and interactions of real-world enterprises onto a conceptual schema.
- The E-R data model perceives the real world as consisting of basic objects, called entities, and relationships among these objects.
- The E-R model has an associated diagrammatic representation, the E-R diagram. E-R diagram can express the overall logical structure of a database graphically.
- Three principal concepts:
  - Entity sets
  - Attributes
  - Relationship sets
- We will consider *Chen Model Notations* for our E-R Diagram.

E-R Diagram ≫Components

- Entity: It is a real-world object about which we collect data. In other words, it is a definable thing that can have data stored about it.
- Entity Set: An entity set is a set of entities of the same type that share the same properties or attributes.
- Attributes: Attributes are descriptive properties of the entities in the entity set. Each entity has its own value for each attribute.
- Relationship: It is an association among several entities.
- Relationship Set: It is a set of relationships of the same type.

E-R Diagram ≫Components

Relation:
Borrower

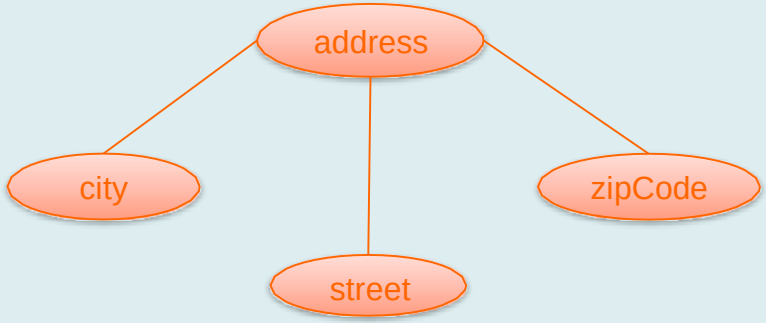


E-R Diagram ≫Attributes

Simple – Attribute value is atomic and can't be divided into subparts.



Composite – Attributes that can be divided into subparts i.e. other attributes.



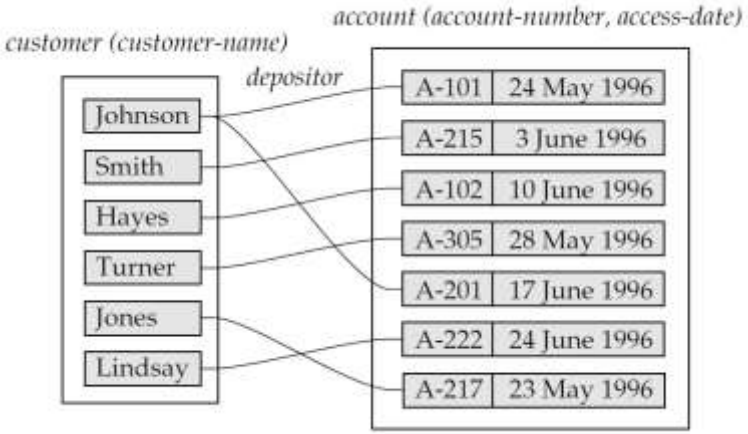
Derived – Attributes that can be calculated from other related attributes.



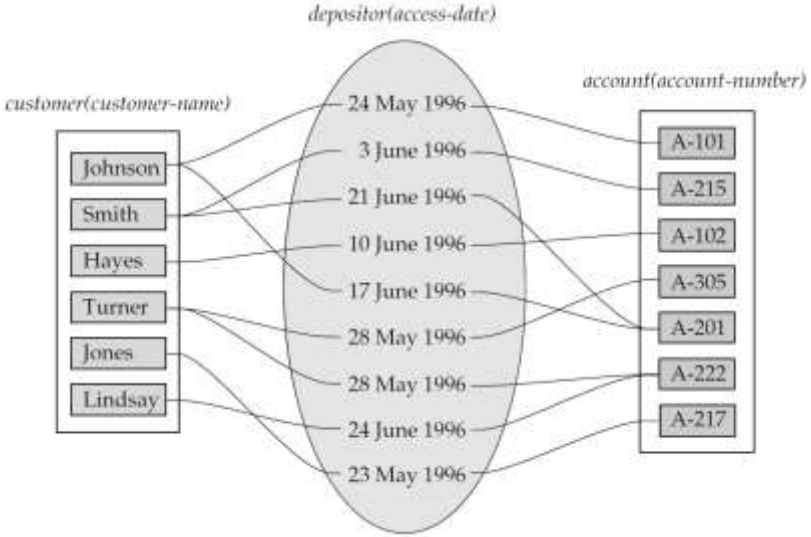
Multivalued – Attributes having more than one values (a set of values) for a specific entity.



E-R Diagram ≫Relationship Attributes



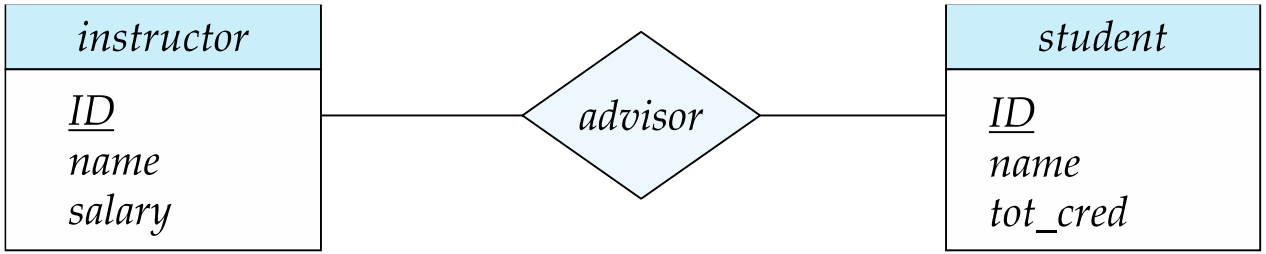
One-to-Many Relationship Set



Many-to-Many Relationship Set

E-R Diagram ≫Relationship Attributes

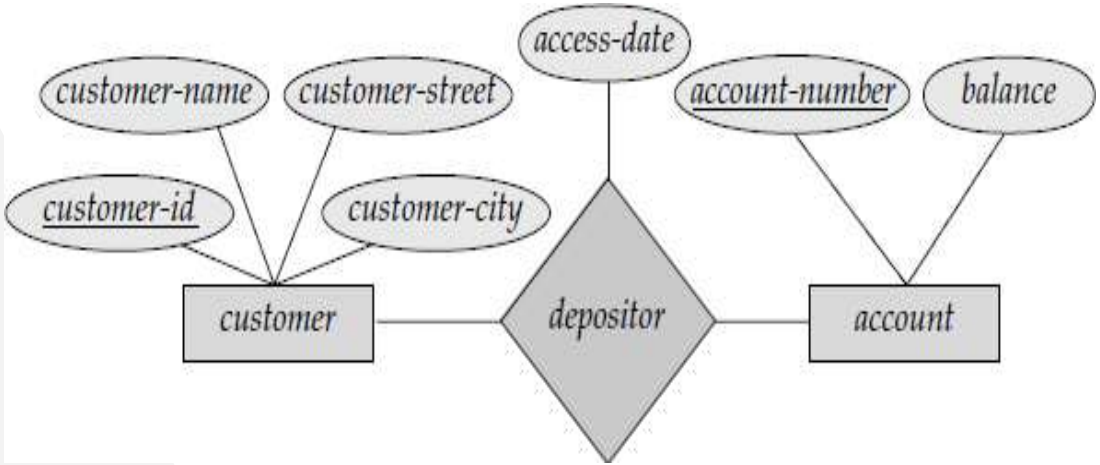
- Diamonds represent relationship sets.



E-R Diagram ≫Relationship Attributes

A **descriptive attribute** is tied to a relationship and not to any individual entity.
Relationship: *"Borrow"* between Customer and Loan.
Descriptive Attribute: **borrow_date** (the date when the customer borrowed the loan).
Relationship: *"WorksOn"* between Employee and Project.
Descriptive Attribute: hours_worked (indicating the number of hours an employee worked on a specific project).
Cannot Belong to Entities:
These attributes cannot logically belong to either of the connected entities individually. For example, borrow_date makes sense only in the context of the relationship between Customer and Loan.

E-R Diagram ≫Relationship Attributes



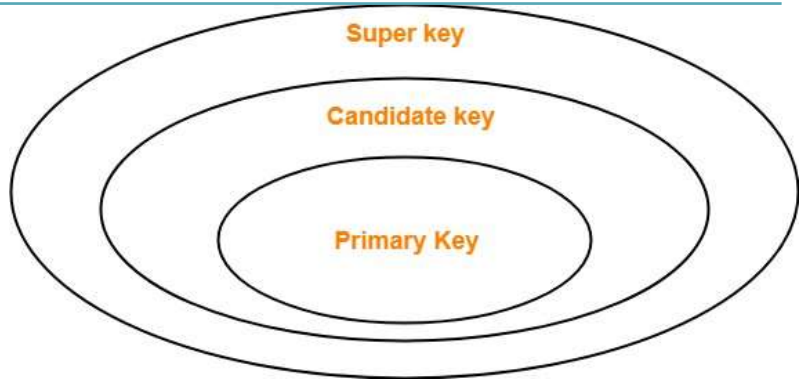
E-R Diagram ≫Keys

Superkey – A set of one or more attributes that, taken collectively, allow us to identify uniquely an entity in the entity set.

Candidate key – A candidate key is a minimal set of attributes that uniquely identifies a row in a table, meaning no other attribute can be removed from it without losing its ability to uniquely identify that row.

Primary key – The candidate key chosen to be used for identifying entities and accessing records.

Key attribute – An attribute that is part of a (primary) key.



Partial key/Discriminator – a discriminator is an attribute (or a set of attributes) in a weak entity set that, combined with the primary key of a strong entity set, uniquely identifies a record. Also known as a partial key, it distinguishes between records within that weak entity set, as the weak entity set does not have enough attributes to be a primary key on its own.



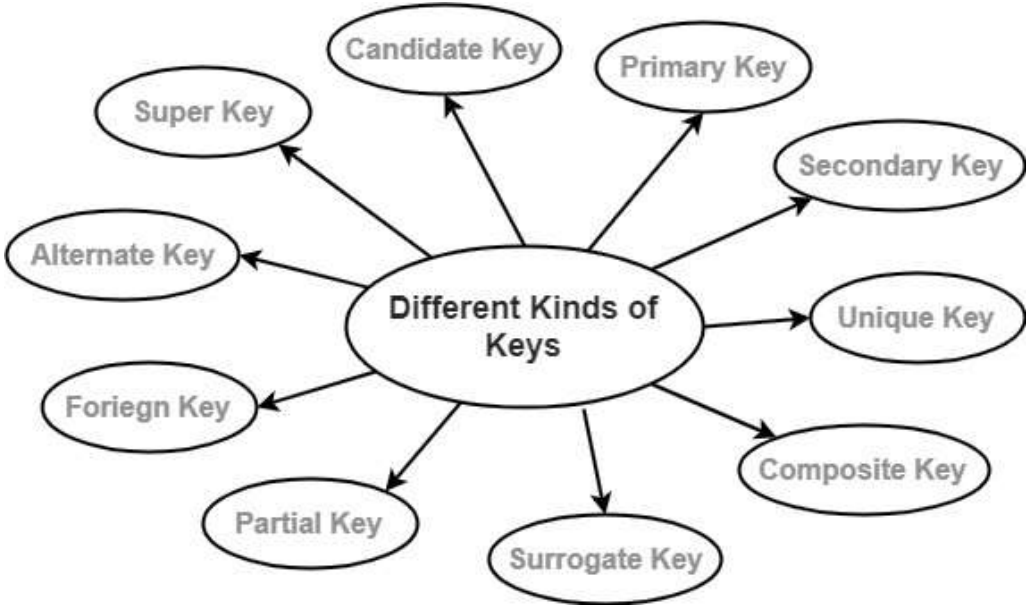


Relational Database Keys

Composite key – A key that is composed of more than one attribute.

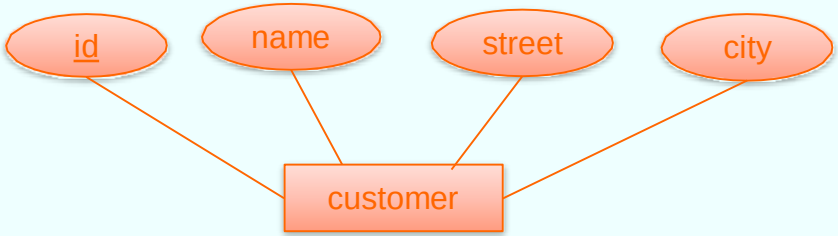
Foreign key – An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null.

Foreign keys are used to ensure referential integrity, the condition in which every foreign key entry must either be null or a valid value in the primary key of the related table.

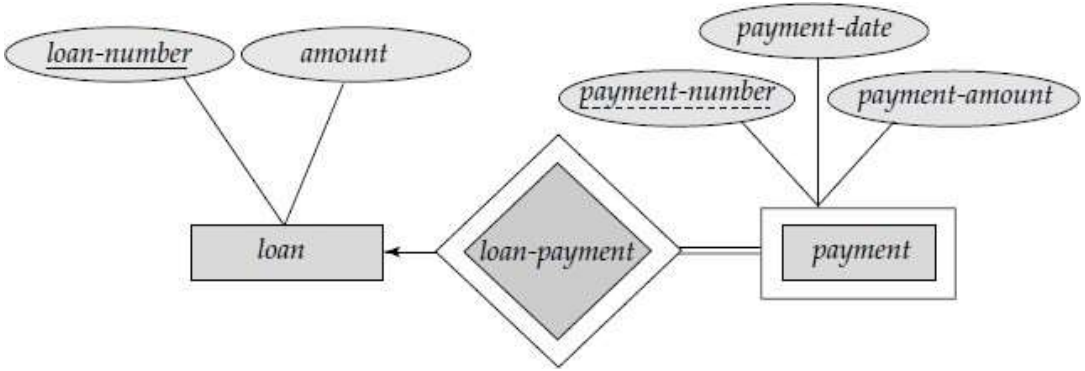


E-R Diagram ≫Entity Sets

Strong Entity Set – A Strong Entity is a type of entity that has a key Attribute. Strong Entity does not depend on other Entity in the Schema. It has a primary key, that helps in identifying it uniquely, and it is represented by a rectangle.



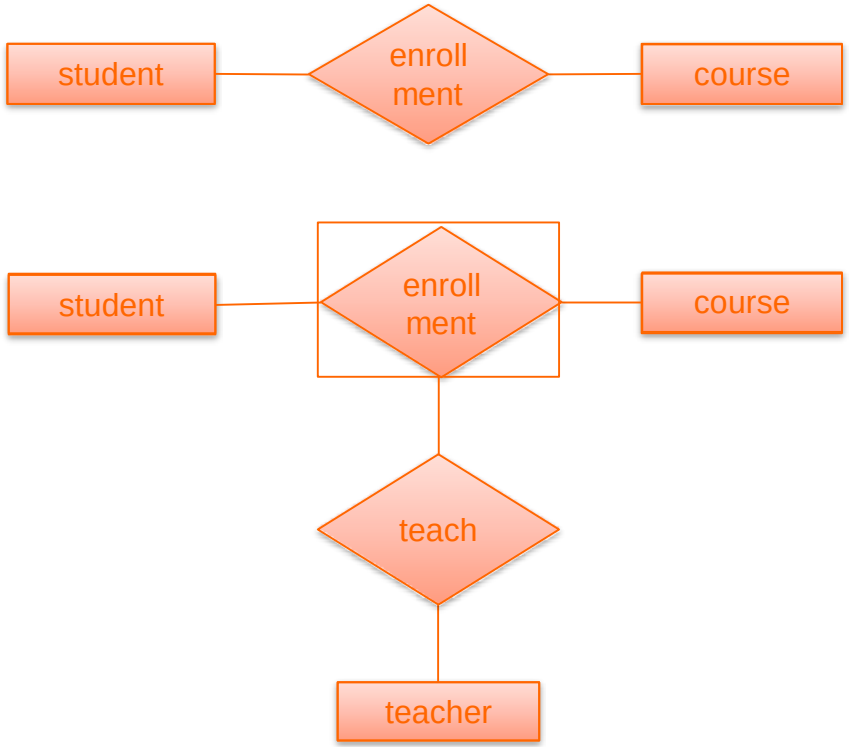
Weak Entity Set – It is one whose existence is dependent on another (identifying) entity set.
The primary key attributes of the identifying entity set, along with the discriminator (partial key) attributes of its own are used to uniquely identify each weak entity.
Here,
Partial key is payment-number . But it is not enough to uniquely identify tuples. It depends on Foreign key **loan-number.** A weak entity type is **represented by a Double Rectangle.**



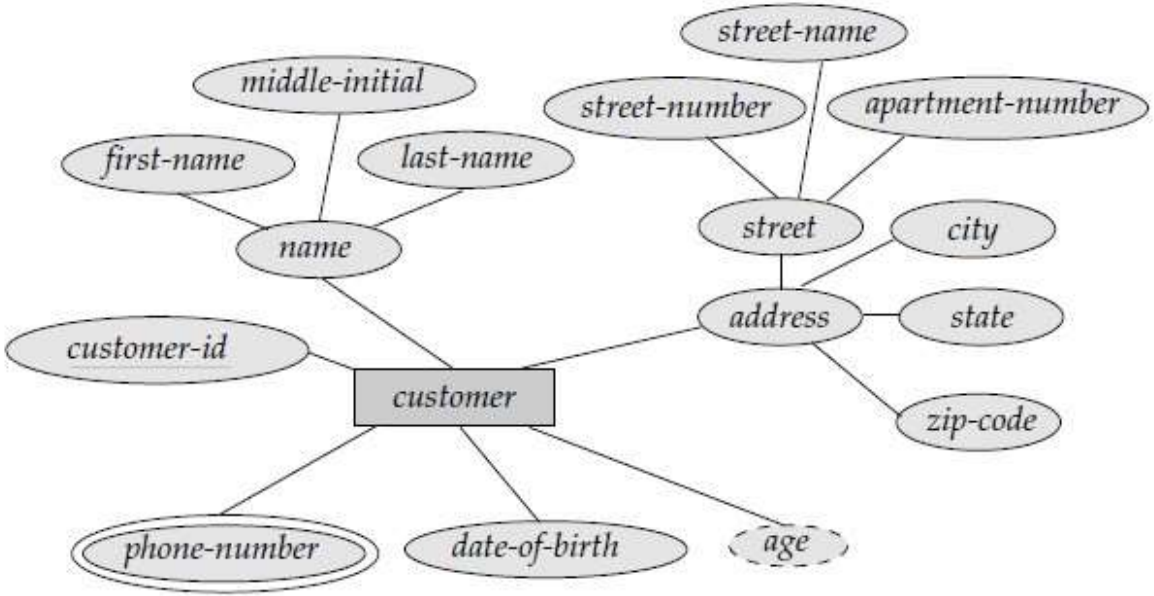
E-R Diagram ≫Entity Sets

Associative Entity Set – This type of entity set is used in a many-to-many relationship.

When you need a relationship to be involved in a relationship then the former relationship is converted to an associative entity.



E-R Diagram ≫Example



E-R Diagram ≫Relationship Sets

Non-Identifying Relationship Set – A non-identifying relationship is a database relationship where a parent entity's primary key is included in the child entity, but not as part of the child's primary key

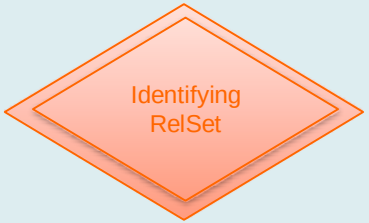
In terms of relationship strength, it is also known as weak relationship.



Identifying Relationship Set – an **identifying relationship** is a connection between two entities where the child entity cannot be uniquely identified without the primary key of the parent entity

The identifying relationship set should not have any descriptive attributes, since any such attributes can instead be associated with the weak entity set.

In terms of relationship strength, it is also known as strong relationship.



E-R Diagram ≫Relationship Sets

Non-Identifying Relationship Example
Entities:
1.Branch:
- •Attributes: branch_id (Primary Key), branch_name.

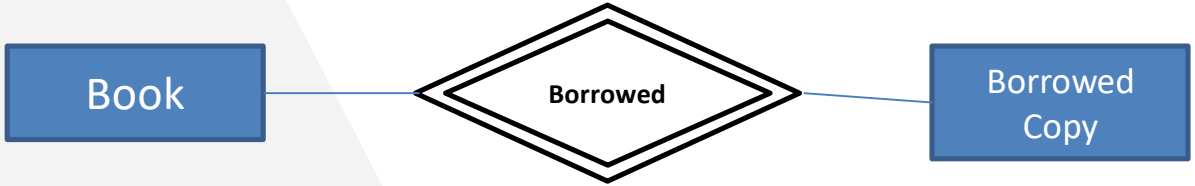
2.Book:
- •Attributes: book_id (Primary Key), title, branch_id (Foreign Key).

Relationship:
Each book is associated with a library branch, but books can exist independently of a branch (e.g., relocated to another branch).
Primary Key:
·Book has its own primary key (book_id), which is independent of Branch.

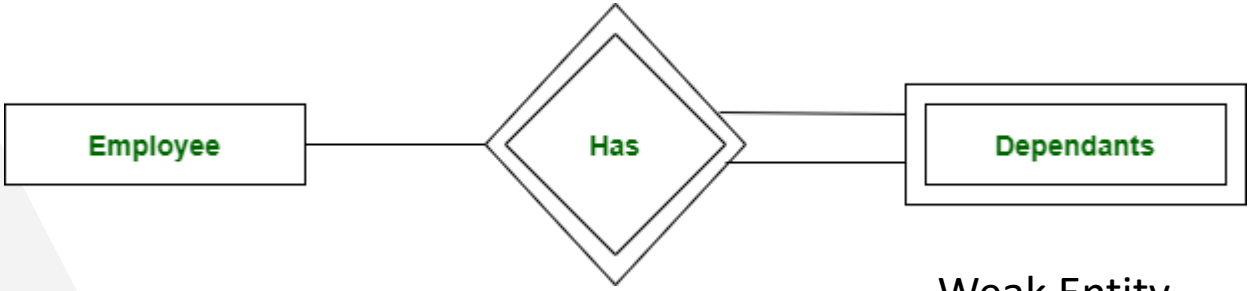


E-R Diagram ≫Relationship Sets

Identifying Relationship Example
Entities:
1.Book:
Attributes: book_id (Primary Key), title.
2.BorrowedCopy:
1. Attributes: copy_id (Partial Key), borrow_date, book_id (Foreign Key).
Relationship:
*Each borrowed copy is uniquely identified by its **copy_id and the book_id** it belongs to. If the book is deleted, the borrowed copy record cannot exist.*
Primary Key:
•BorrowedCopy's primary key is a **composite key**: (book_id, copy_id).
ERD Representation:
•Relationship: *Borrowed* (double-lined diamond between Book and BorrowedCopy).

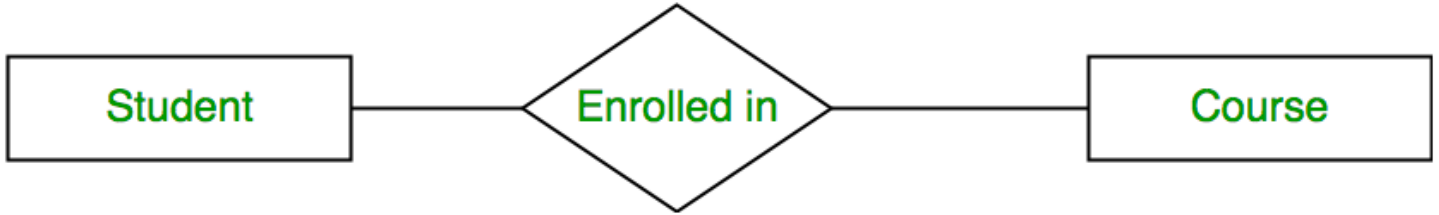


E-R Diagram ≫Relationship Sets



Identifying relationship

Weak Entity

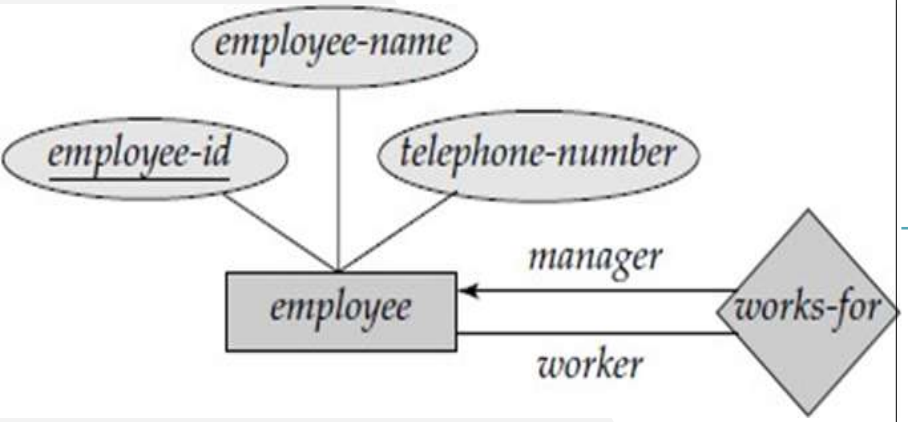


Not Identifying

E-R Diagram ≫Relationship Sets

Recursive Relationship Set –

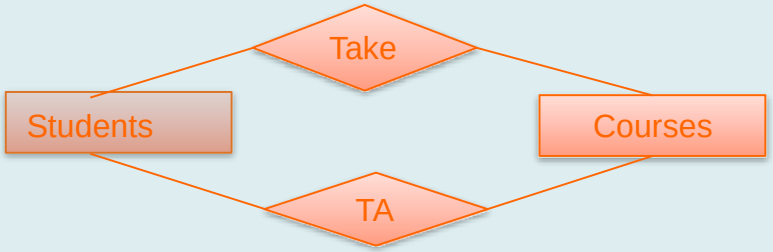
Recursive **relationship** (also known as a **unary relationship**) in a Database Management System (DBMS) is a relationship where the same entity set participates in a relationship with itself, but with different roles for each instance. .



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E-R Diagram ≫Relationship Sets

Parallel Relationship Set – more than one relationships between the same entity sets.

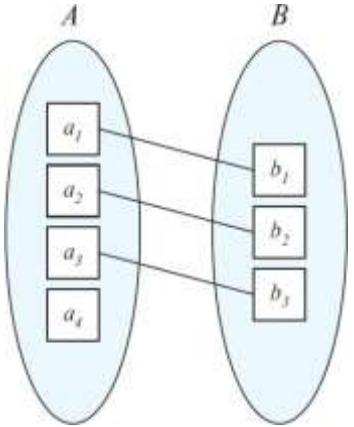


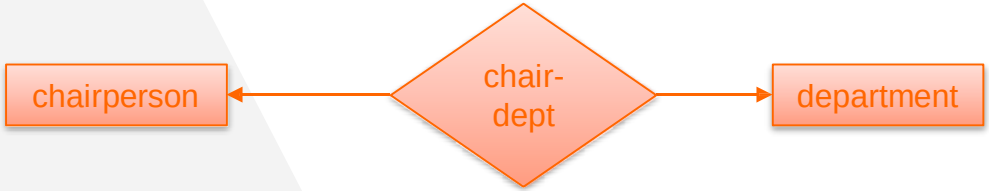
E-R Diagram ≫Mapping Cardinalities

Mapping cardinalities, or Cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.

One-to-one(1:1) – An entity in A is associated with at most one entity in B, and an entity in B is associated with at most one entity in A.

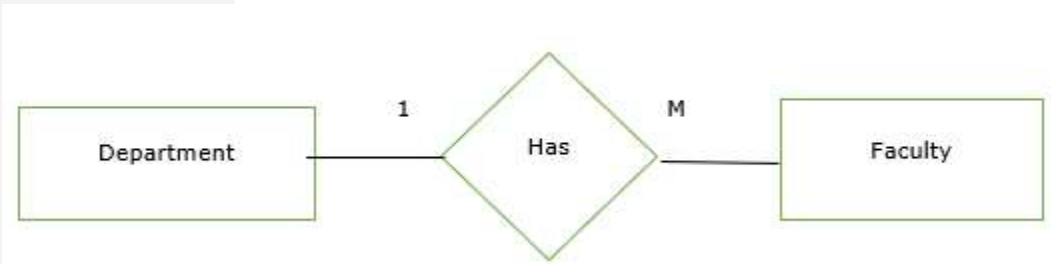


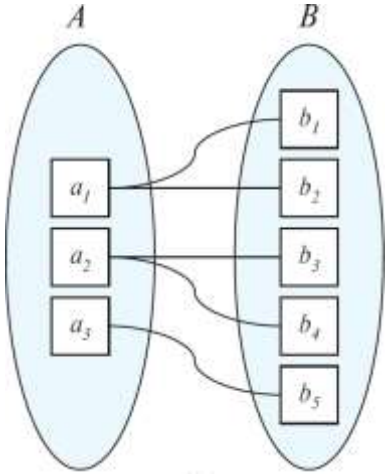




E-R Diagram ≫Mapping Cardinalities

One-to-many (1:N) – An entity in A is associated with any number of entities in B. An entity in B, however, can be associated with at most one entity in A.

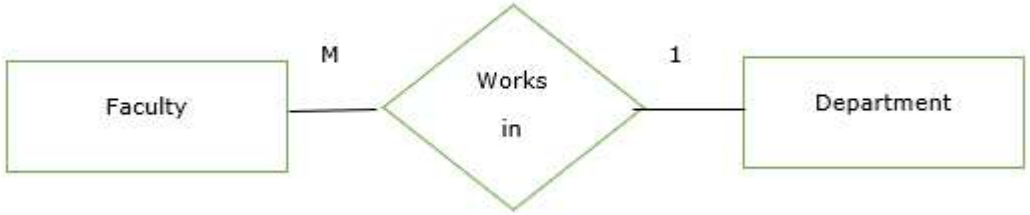


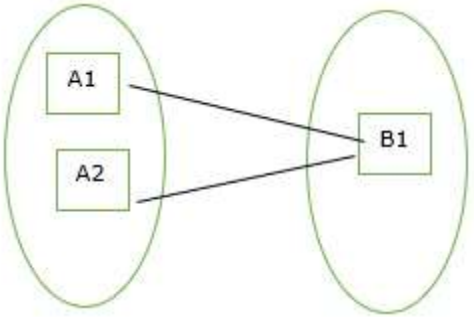




E-R Diagram ≫Mapping Cardinalities

Many-to-One (M:1) – An entity in A is associated with any number of entities in B. An entity in B, however, can be associated with at most one entity in A.

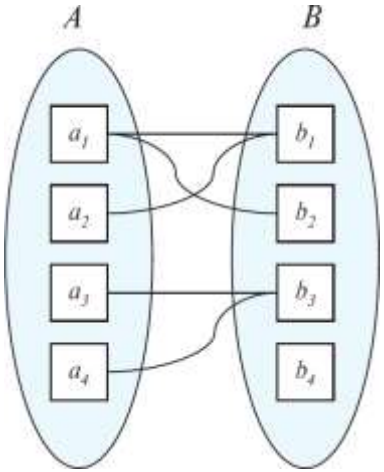


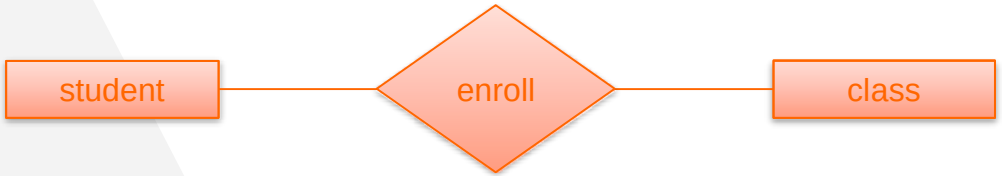


E-R Diagram ≫Mapping Cardinalities

Many-to-many (M:N) – An entity in A is associated with any number of entities in B, and an entity in B is associated with any number of entities in A.







E-R Diagram ≫Participation Constraints

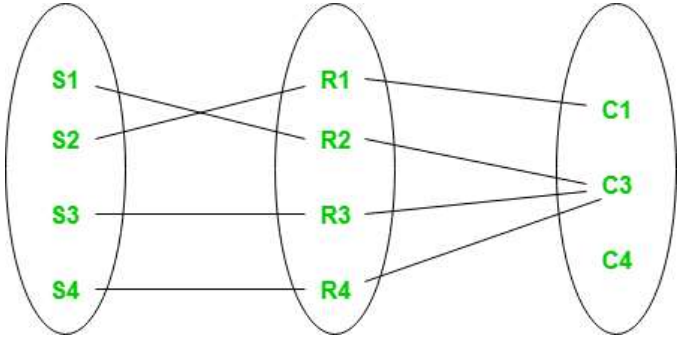
Total participation – Every member of the entity set must participate in the relationship.

Total participation is shown by a double line in the ER diagram.



Partial participation – Not all entities in the set are involved in the relationship. In the previous example not every professor guides student.

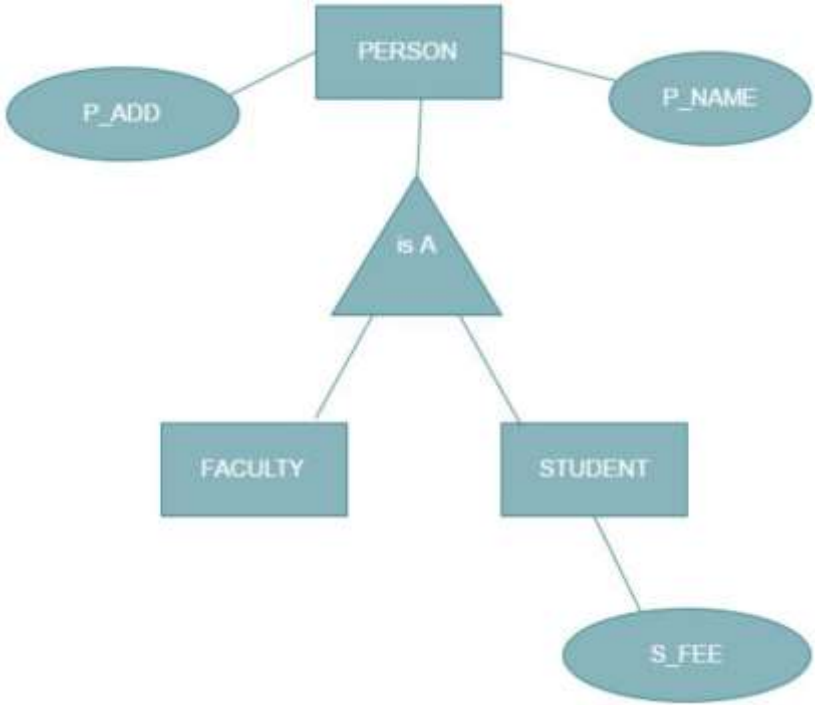




Extended E-R Diagram ≫Generalization

Generalization – It is the process of extracting common properties from a set of entities and create a generalized entity from it.

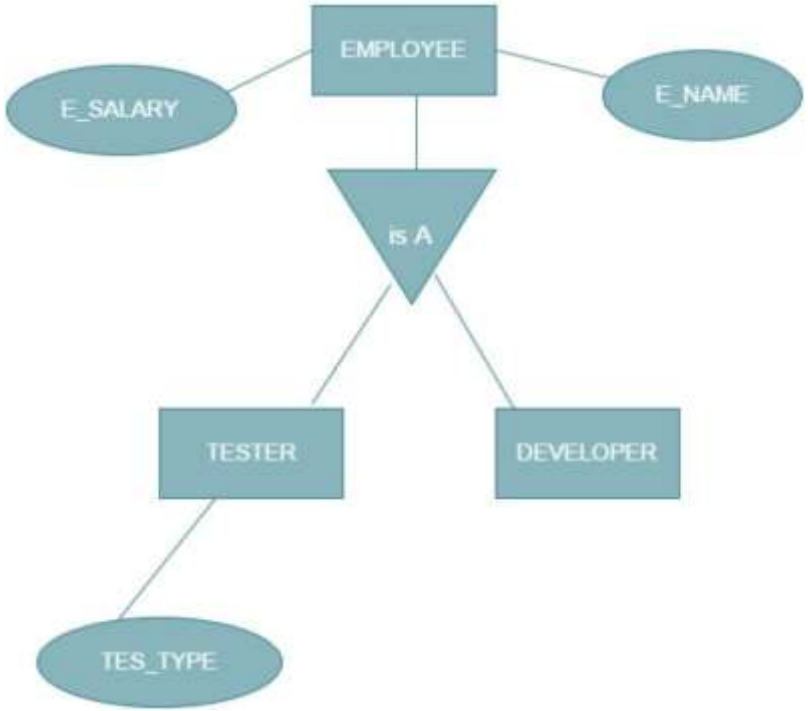
It is a bottom-up approach in which two or more entities can be generalized to a higher level entity if they have some attributes in common.



Extended E-R Diagram ≫Specialization

Specialization – an entity is divided into sub-entities based on their characteristics.

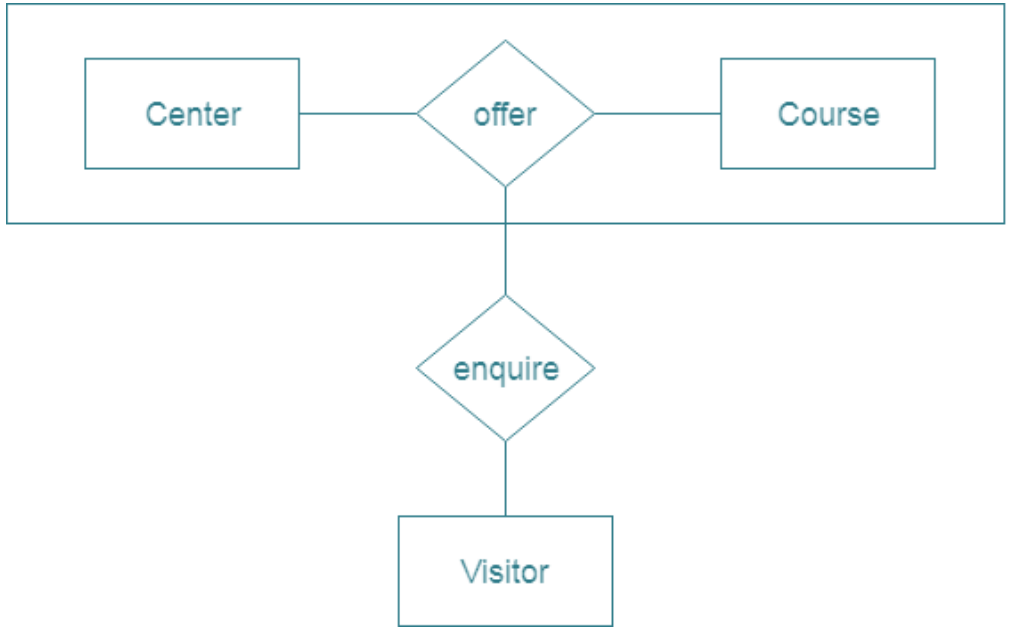
It is a top-down approach where higher level entity s specialized into two or more lower level entities.



Extended E-R Diagram ≫Aggregation

Aggregation –Aggregation is a process when relation between two entities is treated as a **single entity**.

In the diagram above, the relationship between **Center** and **Course** together, is acting as an Entity, which is in relationship with another entity **Visitor**. Now in real world, if a Visitor or a Student visits a Coaching Center, he/she will never enquire about the center only or just about the course, rather he/she will ask enquire about both.



Draw ERD guide

1. **Identify the Entities** -Every entity should be represented by a **rectangle** in the diagram
2. **Define Attributes for Each Entity**-
   - Identify the attributes that need to be stored for each entity.
   - **Primary Key (PK)**: Mark the unique identifier for each entity (usually underlined in ERD). Composite type, derived type
3. **Define Relationships Between Entities** - Draw a **diamond** shape to represent the relationship, and connect it to the related entities.
4. **Determine Cardinality (Multiplicity) of Relationships** –
   - **One-to-One (1:1)**
   - **One-to-Many (1:N)**
   - **Many-to-Many (M:N)**

   Use **crow's feet** to represent "many" and a **straight line** to represent "one".

5. **Identify Weak Entities (If Any)** - A weak entity cannot exist without a parent entity. Draw a **double rectangle** to represent a weak entity.
6. **Identifying Relationship**
    - The child entity's primary key **includes** the primary key of the parent entity as part of its own primary key.The child entity **depends on** the parent for its existence. Without the parent, the child cannot exist.
    - Represented by a **double diamond** in ER diagrams.

7. **Non-Identifying Relationship**
   - **How to Identify:** The child entity's primary key is **independent** of the parent entity's primary key.

     The child entity can exist **without** the parent entity.

   How to Represent:

   Represented by a **single diamond** in ER diagrams.

E-R Diagram ≫Practice 1 (Banking Enterprise)

- The bank is organized into branches. Each branch is located in a particular city and is identified by *a unique name*. The bank monitors the assets of each branch.
- Bank employees are identified by their *employee-id* values. The bank administration stores the name and telephone number of each employee, the names of the employee's dependents, and the *employee-id* number of the employee's manager. The bank also keeps track of the employee's start date and, thus, length of employment.
- Bank customers are identified by their *customer-id* values. The bank stores each customer's name, and the street and city where the customer lives. Customers may have accounts and can take out loans. A customer may be associated with a particular banker, who may act as a loan officer or personal banker for that customer.
- Accounts can be held by more than one customer, and a customer can have more than one account. Each account is assigned a unique account number. The bank maintains a record of each account's balance, and the most recent date on which the account was accessed by each customer holding the account.
- A loan originates at a particular branch and can be held by one or more customers. A loan is identified by a unique loan number. For each loan, the bank keeps track of the loan amount and the loan payments. Although a loan payment number does not uniquely identify a particular payment among those for all the bank's loans, a payment number does identify a particular payment for a specific loan. The date and amount are recorded for each payment.

Construct an E-R Diagram for the bank enterprise.

E-R Diagram ≫Practice 2 (Company Database)

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, social security number, address, salary, gender, and birth date. An employee is assigned to one department but may work on several projects, which are not necessarily controlled by the same department. We keep track of the number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee.
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.

Draw the E-R Diagram.

E-R Diagram ≫Practice 3 (Online Book Store Database)

- Every book has a title, isbn, year and price. The store also keeps the author and publisher for any book.
- For authors, the database keeps the name, address and the url of their homepage.
- For publishers, the database keeps the name, address, phone number and the url of their website.
- The store has several warehouses, each of which has a code, address and phone number. The warehouse stocks several books. A book may be stocked at multiple warehouses. (In previous sentence, we are not referring to a particular copy of the book. Consider for example "the complete book" for our course. This book may be stocked at multiple warehouses.) The database records the number of copies of a book stocked at various warehouses.
- The bookstore keeps the name, address, email-id, and phone number of its customers.
- A customer owns several shopping basket. A shopping basket is identified by a basketID and contains several books. Some shopping baskets may contain more than one copy of same book. The database records the number of copies of each book in any shopping basket.

Design an ER diagram for such a bookstore.

E-R Diagram ≫Practice 4 (Blog Management System)

Construct an ER diagram according to the details provided below:

- The blog management system maintains each blog writer's unique email, name, encoded password, gender, date of birth, join date, length of employment and address. The address field consists of three components: postal code, city and street name. Every blog writer is hired by only one blog writer as his employer. The visitors may follow more than one blog writer.
- Each of the blogs has blog id, details, date, time and category. One blog may fall under several categories. Here blog id cannot uniquely identify a specific blog but blog id together with blog writer email can uniquely identify a specific blog. A blog writer may publish several blogs.
- The visitors can see all the blogs published by writers. Each visitor can comment more than one time in each blog and the system maintains the date, time, content of that blog comment. Visitors can also hit reactions (like/dislike) in more than one blogs. The system maintains unique visitor email, visitor name and encrypted password for each visitor.

E-R Diagram ≫Practice 5 (League of Villains)

Consider the following Scenario:

The League of Villains is getting more and more prominent. To reduce the dominance of the villains,

hero society is thinking of positioning heroes in regions. However they first need to design a database

with the following conditions:

- Heroes have a unique name, a power, strength level and a rank. Villains also have properties similar to heroes. Also, each villains may have a boss who is also a villain himself.
- A hero is responsible for a certain region. A region has latitude, longitude, and population. A region can be terrorized by multiple villains. However, one villain does not terrorize more than a region.
- Heroes and villains often fight. Hero society needs to keep track of the dates and results of thesefights. A hero and a villain may fight multiple times.

Design an ERD for the scenario.

E-R Diagram ≫Practice 6 (Course Mgmt. System)

Consider the following scenario for Course Management System:

- The university stores all the student information that includes a unique student id, name, date of birth, address. Address is stored in the following three sections: street name, postal code and city.
- Each course is identified by its unique course code, trimester, course name and course content. One course is conducted by several faculties and a student can choose several courses in a single trimester.
- The university identifies each faculty through their unique employee id. Each faculty also has a particular room no, name, contact no, email and join date. Each faculty is associated with anonymous reviews of students. Each review consists of a review id, review details and datetime. The review id can't identify a particular review among all the reviews but the review id for a specific faculty can identify a particular review. Each student is also assigned to a specific faculty for advising purpose.

Design an ER Diagram for the scenario that satisfies all the given requirements.

E-R Diagram ≫Practice 7 (Job Portal System)

Consider the following scenario for a Job portal:

- The job portal manages each user's profile where each user is represented by his or her name, unique email, address, date of birth, educational qualifications, job position and job experience. Both recruiters and job seekers are user and one recruiter can recruit more than one job seeker.
- The recruiters can post job openings to the job portal. The job portal keeps track of each posts date, time, contents, no of viewers. The system also keeps track of which job seeker is interested in which job posts.
- Each of the offices is maintained by their unique office id, name, address and contact no. Every user belongs to some offices as a recruiter or, an employee or, a follower. Each person can provide his or her anonymous review of offices. Each of the review contains review id, date, time, review details. The review id cannot uniquely identify a particular review but a review id together with an office id can uniquely identify a particular review.

Design and ER Diagram for the scenario that satisfies all the given requirements.